

L08 — Speech Synthesis I

Text analysis, prosody, diphone and unit selection

Study summary generated from lecture slides. ETH AI for Digital Characters.

TTS pipeline overview

- Stage 1: Text Analysis (normalization, phonetics, prosody)
- Stage 2: Waveform Synthesis (concatenative or parametric)

Text normalization

- Identify tokens -> chunk sentences -> classify types -> expand to words
- Handles: abbreviations, numbers, punctuation disambiguation
- Homographs: live/li:v/ vs /laiv/ via POS tagging + dictionary

Phonetic analysis

- CMUdict: ~127K words, 39 phonemes, stress markers 0/1/2
- G2P: ML classifier for out-of-vocabulary words
- Names: ~20% of tokens; morphology and rhyme rules

Prosodic analysis

- Prominence/accent: which syllable is stressed in context
- Boundaries: intonation phrase (||), intermediate phrase (|)
- Duration and F0 (fundamental frequency) per phone
- Stress = lexical (fixed in word); accent = context-dependent
- Prominence classes: unaccented, reduced, pitch accent, nuclear accent
- ToBI: H* (peak), L* (low), L-H% (continuation rise)

Diphone synthesis

- Diphone: half of each adjacent phone - captures coarticulation
- Record -> segment -> concatenate with Hanning window -> prosody modify
- Epoch labeling: vocal fold cycle markers (Praat)

Unit selection synthesis

- Larger units from ~10h recorded speech; minimal signal processing
- Target cost $T(s_t, u_j)$: match F0, stress, duration, position
- Join cost $J(u_j, u_{j+1})$: smoothness at boundaries (=0 if consecutive in DB)
- Select sequence minimizing sum of target + join costs